PU Learning (H05N0a)

Report: Homework

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1 Initial Analysis Questions

Question 1 SEM should perform better on the first dataset. This is because of its labeling mechanism which is SAR. This biased labeling mechanism creates two clusters from the positive data and labels examples from one of those clusters. This method makes the labeled and unlabeled data more likely separable. Depending on how these clusters are created the smoothness assumption can hold too. These are two assumptions on which two-steps techniques such as SEM rely on.

On the second dataset MLR should be more suited instead. In the second dataset the labeled examples are selected with the SCAR assumption from the positive examples. MLR incorporates the class prior making use of the label frequency c in a post processing phase. This way, the unlabeled data can be considered as negative and at the end the probabilities can be adjusted with an estimate of c.

Question 2 Both SEM and MLR perform well on the first dataset. They both achieve an F1-score of 0.994. MLR still performs well because the positive subdomain assumption, which says that a region with a high concentration of labeled examples must include only positive examples, holds. The estimate for c is 0.209 when the real c equals 0.2.

On the second dataset MLR performs better than SEM. The F1-scores are 0.599 for SEM and 0.689 for MLR. SEM did not perform well as expected due to the SCAR assumption and separability is also not guaranteed. MLR did better but not good either. The estimated c is 0.105 while the true c is 0.198 and the probabilities that it returns for the test examples being positive are all 1. It is possible that the MLR algorithm got stuck at a local maximum during the gradient ascend.

Question 3 The MSE scores of SEM and MLR on the first dataset are 0.008 and 0.006. These are very good results, it means that getting different classification results from the golden standard classifier and the produced classifiers is rare especially if the larger variations tend to be at the extremes (where the probabilities are almost 0 or 1).

On the second dataset the MSE scores are 0.056 and 0.289 for SEM and MLR respectively. It is interesting to see MLR having a much higher MSE than SEM but still obtaining a better F1 score. The extremely high MSE of MLR is due to all the probabilities in the test case being equal to 1.

Question 4 With the first dataset where SAR, separability and smoothness hold SEM has no problems and MLR also performs well, even if one of its assumptions is SCAR which does not hold because of the positive subdomain assumption. However with the second dataset where SCAR holds both methods perform significantly worse.

2 Research Questions

Question 1 Is it possible to alleviate the absence of separability in a dataset where SCAR holds with SEM by making use of different classifiers?

In the implementation of SEM the default classifier that is used during training is a logistic regression. This classifier struggles to fit inseparable data and often does not converge during EM iterations. This lack of convergence during the *clf.fit()* phase also causes the algorithm to take more time. An alternative could be to use Gaussian naive Bayes. Because of the SCAR assumption it seems reasonable to make a Gaussian curve from the labeled examples. In other iterations of the EM process unlabeled examples that are more likely to belong to the positive curve will get higher positive weights while the labeled examples will keep their positive weights on 1 and will not be affected by the negative curve. An experiment on the second dataset gives an F1-score of 0.657 which is higher than SEM with a logistic regression classifier. Step 1 also converges after 49 iterations while with logistic regression got stopped after 100 iterations. Step 1 generates an N set of only 17 examples instead of 503. Step 2 only makes 1 iteration which means that the final classifier is made with those 17 examples as negatives and the labeled as positives.

Question 2 What are the effects of changing the spy proportion $s\%$ and the noise level $l\%$?

Changing the spy proportion is not going to change much as long as enough data is used. The main problem that can occur when changing the spy proportion is that the spies that are being used are very positive examples. This can especially occur if the given spy proportion is too low and not enough spies are being used.

Augmenting the spy proportion does not create problems because of the threshold determined by the noise level. Adding a spy that is situated higher than the threshold could simply make the threshold shift to one spy higher after having sorted all spies by their probabilities. Adding a spy that is lower than that threshold is simply going to possibly lower the threshold. Since this selection happens randomly the threshold is not expected to shift too much. However having less spies implies having to use less positive labeled examples during the EM in step 1, this could slow down the convergence of the EM.

The noise level is not going to affect the result too much either. If the threshold would be situated too high and wrongfully put positive examples in the probably negative set, the weights of these examples would quickly change during the EM in step 2. This is because the weights of the examples in the probably negative set are not being reset after each EM iteration like with the labeled examples.

Question 3 Looking at what happens with the b and w during iterations of MLR can be interesting. You notice that for w of the first dataset the values oscillate. This means that it could be good to chose to do more iterations with a smaller learning rate. The values of the w of the second dataset converge very rapidly to (maybe) a local maximum. If that is the case, starting with different w could possibly lead to another solution.

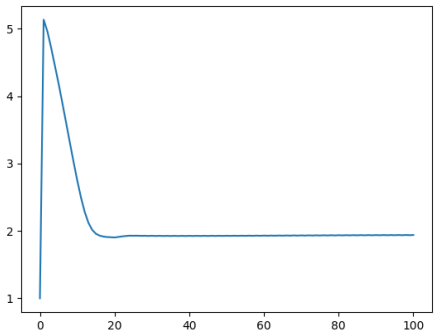


Figure 1: How b changes for MLR on the first dataset. The y-axis represents its value and the x-axis represents the number of iterations

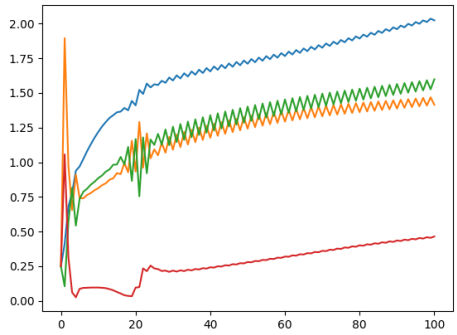


Figure 2: How the w changes for MLR on the first dataset. The y-axis represents the values of its components and the x-axis represents the number of iterations

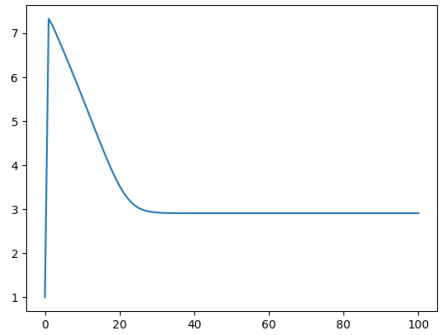


Figure 3: How b changes for MLR on the second dataset. The y-axis represents its value and the x-axis represents the number of iterations

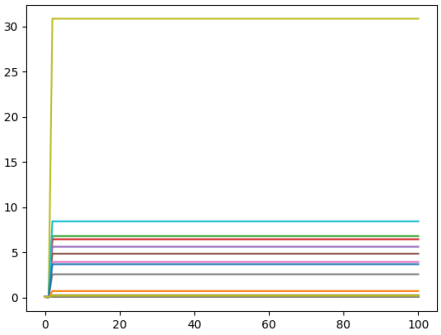


Figure 4: How the w changes for MLR on the first dataset. The y-axis represents the values of its components and the x-axis represents the number of iterations